

Anjitha Mekkayil Anand

Senior .NET Engineer | Agentic AI · RAG · LLM Evaluation | C# · Azure · MCP

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Profile

Senior .NET engineer, ten years in .NET and twelve in software, across enterprise platforms and cloud systems. Current work sits where enterprise .NET meets agentic AI: C#, .NET 10, Blazor, SQL Server and Azure on one side; RAG pipelines, MCP tool surfaces and reasoning agents on the other.

Recent work: 279 merged pull requests and 812 pull requests reviewed across a multi-tenant, multi-language Nordic energy platform. Mathesis, a four-agent system with a custom MCP tool server and Azure AI Search grounded retrieval, for the Microsoft Agents League. EG ClarityHub for Talent Accelerator 2026 (Runner-Up).

Open to senior .NET and architect-track roles, remote.

Technical Skills

Languages & Frameworks: C#, .NET 10, ASP.NET Core, Blazor, Web API, Entity Framework Core, KnockoutJS, JavaScript, HTML, CSS, LINQ, ADO.NET

AI & Agentic Systems (public projects, 2026; Claude Code daily at work): RAG pipelines with grounded citation, LLM evaluation (retrieval metrics, LLM-as-judge), multi-provider routing with fallback, MCP servers, Claude API, OpenAI API, GitHub Copilot

Cloud & Infrastructure: Microsoft Azure: IoT Hub, AI Search, Functions, Cosmos DB, SQL Server, Web Apps, Azure AD, Event Hubs, Azure DevOps (CI/CD) · Docker · Observability: OpenTelemetry, Azure Monitor, Application Insights

Databases & Tools: MS SQL Server (query optimization, index design, performance tuning), SQLite, SSMS, Visual Studio, Git

Patterns & Practices: Clean Architecture, CQRS, Repository, Decorator, Modular Monolith, Multi-tenant Architecture, Microservices, OAuth 2.0 / OpenID Connect, Agile / Scrum

Professional Experience

Senior Software Developer

Aug 2023 – Present

EG India, Mangaluru

Enterprise energy management platform · .NET / Blazor / SQL Server · Nordic utility clients

- Delivered features, architectural improvements and production fixes across five modules of a large, multi-tenant, multi-language .NET/Blazor/SQL Server modular monolith serving Nordic utility operators: billing workflows, metering data pipelines and market protocol integrations, across **279 merged pull requests**
- Built **EG ClarityHub**, a full-stack AI-integrated CRM platform, in .NET 10 and Blazor for Talent Accelerator 2026, one of 26 across EG's Industrials & Trade division. Presented live to the Exec VP, CTO and SVP panel; panel response: *"We'd like this in production today."* **Runner-Up** across development, product, support and commercial evaluation
- Reviewed **812 pull requests** on colleagues' code as the cross-team review path, accountable for work delegated to junior engineers
- Root-cause analysis across frontend, backend and database layers; SQL Server and EF Core query optimization and index design, including an N+1 pattern reduced from one call per record to two calls total, regardless of record count
- Refactored a 7,000-line front-end module where adding a single status meant hand-editing 17 hardcoded arrays. Replaced them with a three-layer resolution pipeline, surfacing 12 pre-existing bugs including a silently incorrect status-group mapping, and cutting the two largest files by ~75%
- Versioned REST API development for external client integrations
- Applied AI-augmented development practices using Claude Code, MCP tool integration and agentic workflows; established patterns for AI-assisted root-cause analysis and introduced the approach to the team

C# · .NET 10 · Blazor · KnockoutJS · SQL Server · Entity Framework Core · Azure

Senior Software Engineer

Aug 2022 – Jul 2023 · 11 mos

Zowork, Bengaluru (Remote)

- Built the real-time device-to-client event pipeline on Apache Kafka and Azure Event Hubs across distributed systems in .NET Core
- Implemented OAuth 2.0 and OpenID Connect authentication and authorisation with IdentityServer4 across a microservices platform, securing inter-service communication
- Designed and maintained RESTful backend services in .NET Core and C# across a distributed microservices architecture: API design, performance and data integrity for enterprise clients
- Mentored junior developers through code review across the platform's backend services

C# · .NET Core · SQL · Azure Web App · Azure SQL DB · OAuth · Microservices · Kafka · Dapper · MediatR

.NET & Cloud Developer

May 2019 – Aug 2022 · 3 yrs 3 mos

WeMakeloT, Calicut, Kerala

- Designed and deployed an Azure IoT monitoring platform across 2 enterprise clients (IoT Hub, Functions, Cosmos DB, Azure AD, ApplInsights and Power BI dashboards), covering water infrastructure and industrial IoT across rural and commercial deployments
- Migrated a legacy ASP.NET application to .NET Core for a global financial services client, modernising the stack and improving maintainability
- Built .NET Core web APIs and websites hosted on Azure, with CI/CD through Azure DevOps
- Built ESP32 device firmware and web server for remote monitoring and control, bridging device-level work with the cloud platform

C# · .NET Core · ASP.NET · Entity Framework · Azure IoT · Cosmos DB · Azure Functions · Azure AD · SQL · JavaScript

Senior Developer & Team Lead

Apr 2017 – Apr 2019 · 2 yrs

OHO Software Innovations, Cochin

- Converted a live billing product to India's GST regime at launch: learned the new tax rules directly from the legislation, mapped them onto the existing billing model and shipped to 10+ enterprise clients
- Led migration of a legacy FoxPro billing system and its DBF databases to .NET C# and SQL Server, evolving it into an integrated billing and reporting platform
- Led a team of ~6 engineers through the full SDLC, requirements through to production delivery
- Delivered SSRS reports, complex stored procedures and T-SQL for an ERP platform

C# · WinForms · ASP.NET · ADO.NET · SQL · SSRS · JavaScript · jQuery · IIS

.NET & SQL Server Developer

Jan 2016 – Apr 2017 · 1 yr 3 mos

OHO Software Innovations, Cochin

- Built and maintained intranet and ERP applications in ASP.NET and SQL Server
- Developed complex stored procedures, user-defined functions and SSRS reports for data analysis

C# · ASP.NET · ADO.NET · SQL · SSRS · JavaScript · jQuery · IIS

Project Trainee Engineer

Aug 2014 – Jun 2015 · 10 mos

Gadgeon Smart Systems Pvt. Ltd., Thrikkakara

- M.Tech project internship in real-time computing: compared response times across Xenomai, ARM Cortex-A8 and the PRU subsystem on AM335x, then built a real-time data collection library using Programmable Real-time Units, achieving deterministic timing without an RTOS
- Cross-compiled the Linux kernel for AM335x and BeagleBone Black; built a plotting, logging and alert system by embedding Python in C
- Work contributed to two published research papers (ACM WCI'15 and IJERT)

Embedded C · Linux · PRUSS · AM335x

Software Engineer

Sep 2006 – Feb 2008 · 1 yr 6 mos

Infosys, Mangaluru

- Mainframe development for enterprise clients: COBOL, DB2, JCL batch processing, CICS online screens

COBOL · DB2 · JCL · CICS

Personal Projects

Anamnesis: RAG Pipeline with Evaluation Harness

Jul 2026 · github.com/anjitha-mekayil-anand/anamnesis

Retrieval-augmented generation over my LinkedIn posts and Noesis Letters newsletter archive · Grounded cited answers · Multi-provider routing with failover · Measured quality baseline · .NET 10

- Built a RAG service: paragraph-aware chunking with overlap, OpenAI embeddings stored in SQLite, exact cosine top-k retrieval, and grounded answers with inline citations
- Provider router with Polly resilience policies: Claude primary, retry with exponential backoff and per-attempt timeout, automatic failover to OpenAI
- Evaluation harness with a 20-question golden set: retrieval hit-rate@k and MRR, LLM-as-judge answer faithfulness, per-stage latency; results logged per run as a quality trend
- First baseline: hit-rate@5 0.95, MRR 0.90, faithfulness 1.00. The grounding guard declines rather than inventing when retrieval misses

C# · .NET 10 · RAG · Embeddings · SQLite · Claude API · OpenAI API · Polly · xUnit

Mathesis: Multi-Agent Enterprise Learning System

Jun 2026 · Microsoft Agents League hackathon ·

github.com/anjitha-mekayil-anand/Mathesis

Four specialised reasoning agents · Custom MCP tool server · Azure AI Search grounded retrieval · Human-in-the-loop approval gates · .NET 10

- Designed and built a multi-agent enterprise learning system: orchestrator, skills advisor, content curator and assessment designer agents
- Custom MCP tool server exposing learning domain tools to Claude reasoning agents via Model Context Protocol
- Azure AI Search grounded retrieval with semantic ranking; human-in-the-loop approval gates for all agent recommendations

- Built solo in .NET 10 with full unit test coverage; submitted Jun 2026

C# · .NET 10 · MCP · Claude API · Azure AI Search · SQLite · xUnit

Pronoia: Predictive Maintenance AI System 2026 · github.com/anjitha-mekkayil-anand/Pronoia
Multi-phase .NET 10 system: sensor simulator → change detector → MCP server → Claude reasoning agent → Blazor human approval queue

- **Pronoia.Detection:** SensorEvaluator + ChangeDetector with three states (Normal / Drift / Breach); drift on running average catches the pre-breach pattern predictively
- **Pronoia.Mcp:** MCP server with guardrail-by-tool-surface design; the agent can only read and flag for review, never act directly
- Closes a 20-year arc: BTech factory automation (2006) → ACM-published PRUSS system (2015) → AI reasoning agent (2026)

C# · .NET 10 · SQLite · MCP · Claude API · Blazor · xUnit

Certifications

Microsoft Azure AI Engineer Associate (AI-102)	Jun 2025 – Jun 2027
Anthropic Academy: Claude Code · MCP · Agent Skills	2026
GitHub Copilot Certification (GH-300)	Dec 2025 – Dec 2027
Microsoft Applied Skills: Accelerate AI-assisted development by using GitHub Copilot	Oct 2025
Professional Scrum Master I (PSM I)	Jun 2025 · no expiry
Microsoft Azure Developer Associate	Feb 2021 – Feb 2027

Education

Master of Technology (M.Tech.) in Embedded Systems	2015
National Institute of Electronics & Information Technology (NIELIT), Calicut	
<i>Full-time programme, 2013–2015, preceded by a career break for family responsibilities and relocation.</i>	
Bachelor of Technology (B.Tech.) in Applied Electronics & Instrumentation	2006
Government Engineering College, Kozhikode	

Additional Information

Location: Mangaluru, India · Open to remote and relocation for the right opportunity
Work authorisation: Indian national · Eligible to apply for relevant work visas
Languages: English (Professional) · Malayalam (Native) · Hindi (Conversational)

Publications

Using PRUSS for Real-Time Applications on Beaglebone Black	
WCI '15, Proceedings of the Third International Symposium on Women in Computing and Informatics · Cited 11× (Google Scholar)	
https://dl.acm.org/doi/10.1145/2791405.2791408	
Real-time Application in Linux using PRUSS	
International Journal of Engineering Research & Technology · Cited 2× (Google Scholar)	
https://www.ijert.org/research/real-time-application-in-linux-using-pruss-IJERTV4IS050971.pdf	